Concentration

Check Point

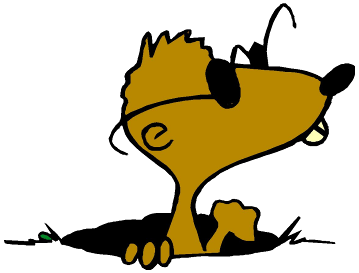
Please return to the website guide!

Concentration

- A solution is a homogeneous mixture of one substance (the solute) dissolved in another substance (the solvent).
- Concentration is a ratio of the amount of solute to the amount of solvent.

Units of Concentration

- Molarity (M) is the most common unit of concentration
- Molarity is an expression of moles/Liter of the solute.
- We will also learn about percentage and very small concentrations, which are useful in certain situations.



Molarity

Molarity is one way to measure the concentration of a solution.

$$\text{Molarity } (M) = \frac{\text{moles of solute}}{\text{volume of solution in liters}}$$

A 1.00 molar (1.00 M) solution contains 1.00 mol solute in every 1 liter of solution.

Units of molarity are: mol/L = M


No one's ever too old for chemistry....
Hey son, your father told me that you needed a 24 molar solution at school .
Yejoon Song
-mak

Molarity (can use c or M for symbol)

$$M = \frac{n}{V}$$

- Where M is the concentration in mol/L. This unit can also be expressed with a capital “M”. So, M can refer to both the variable of molarity and the unit of measurement! Note that the textbook uses different variables…be careful!

Molarity Example

- **Example 1:** What mass of NaCl is needed to make 250 mL of a 0.850 M solution?

- Given:

M = 0.850 mol/L

V = 250 mL = 0.250 L *(convert so that units are consistent)*

Molarity Example

$M = n/V$ $\rightarrow$ rearrange to solve for n, then you can convert n to mass

- $n = MV = (0.850 \text{ mol/L})(0.250 \text{ L}) = 0.212 \text{ mol}$
-
- $m = nMM$ *(don't confuse molar mass (MM) with molarity (M))*
- $m = (0.212 \text{ mol})(58.5 \text{ g/mol}) = 12.4 \text{ g}$
- Therefore, 12.4 g of NaCl is needed to make 250 mL of a 0.850 M NaCl solution

Diluting Solutions

- Often once you have made a stock solution, you need to dilute it to a working concentration.
- To determine how to dilute the stock solution, use the dilution equation:

$$M_1V_1 = M_2V_2$$

M_1 – concentration of stock
M_2 - concentration of diluted solution
V_1 – volume needed of stock
V_2 – final volume of dilution

- NOTE: Dilution does NOT change the number of moles solute in the original volume

Diluting Solutions

Example:

- What volume of 16.0 M nitric acid solution is needed to prepare 500 mL of a 0.500 M nitric acid solution?

Given:

$M_1 = 16.0 \text{ mol/L}$

$M_2 = 0.500 \text{ mol/L}$

$V_1 = ?$

$V_2 = 500 \text{ mL} = 0.500 \text{ L}$

$$M_1 V_1 = M_2 V_2$$

Diluting Solutions

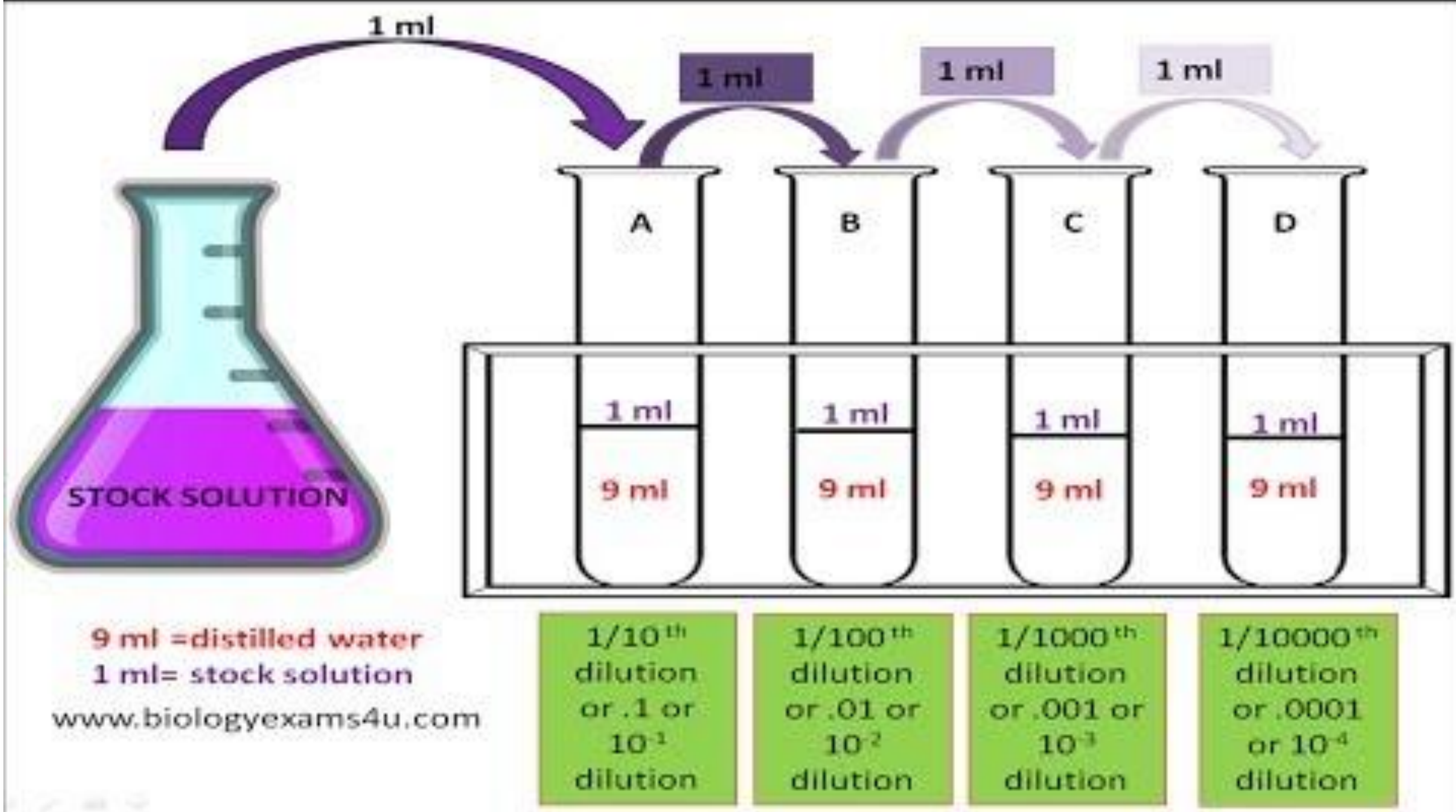
$$V_1 = \frac{M_2 V_2}{M_1}$$

$$V_1 = \frac{(0.500\text{mol/L})(0.500\text{L})}{(16.0\text{mol/L})}$$

$$V_1 = 0.015625\text{L} = 15.6\text{mL}$$

Therefore, to make desired solution, 15.6 mL of 16 .0 M nitric acid solution is used

Next up: Serial Dilution


1 ml
1 ml
1 ml
1 ml
A
B
C
D
1 ml
9 ml
1 ml
9 ml
1 ml
9 ml
1 ml
9 ml
STOCK SOLUTION
9 ml =distilled water
1 ml= stock solution
www.biologyexams4u.com
1/10th dilution or .1 or 10-1 dilution
1/100th dilution or .01 or 10-2 dilution
1/1000th dilution or .001 or 10-3 dilution
1/10000th dilution or .0001 or 10-4 dilution

Diluting Solutions

Unfortunately, due to Adobe Flash discontinuation, this isn't avail

Dilutions tutorial

Takes several minutes to load

Concentrations in Consumer Products

- These concentrations are typically found on the product label.
- Used because you don't have to be familiar with moles.
- Eamples:
  - hair dyes, peroxides
  - medicines like topical creams
  - rubbing alcohol
  - vinegar for food consumption.

% Concentration

- $$\%V/V = \frac{\text{volume solute}}{\text{volume solution}} \times 100\%$$
- $$\%W/V = \frac{\text{mass solute}}{\text{volume solution}} \times 100\%$$
- $$\%W/W = \frac{\text{mass solute}}{\text{mass solution}} \times 100\%$$

- typically the volumes are in mL
- % by mass – often used in medicated creams
- % by volume – often used in rubbing alcohol, vinegar

Concentration Activity

Visit each station and use the calculation given on the sheet to practice determining these percentages from the consumer products.

You can use the rest of the slides in this presentation to further your understanding if needed.

Solution Percentages

This method of determining concentration is common in industrial applications and are typically used to quickly prepare solutions.

- Percent volume (%V/V)

$$\% \text{ volume} = \frac{\text{volume solute (ml)}}{\text{volume solution (ml)}} \times 100$$

Solution = solvent + solute

- What is the % V/V of a solution prepared by adding 10 mL of solute to 45 mL of solvent
- = 10/(10 + 45) x 100% = 18.2 % V/V *(include this term in your answer)*

Solution Percentages

- Percent mass per volume (%W/V)

$$\% \text{ mass} = \frac{\text{mass solute (g)}}{\text{volume solution (ml)}} \times 100$$

Note: given the mass of water is 1 gm/ml, these are essentially the same units (g per g) as water is the most common solvent

Solution = solvent volume as solute is assumed to be negligible

- What is the % W/V of a solution prepared by adding 10 g of solute to 45 mL of solvent
- % = 10/45 x 100% = 22.2 % W/V *(include this term in your answer)*

Solution Percentages

- Percent mass per mass (%W/W)

$$\% \text{ mass} = \frac{\text{mass solute (g)}}{\text{mass solution (ml)}} \times 100$$

Assume water density to be 1.0g/mL

$\boxed{\text{Solution} = \text{mass solvent} + \text{mass solute}}$

- What is the % W/W of a solution prepared by adding 10 g of solute to 45 mL of water

= 10/(10 + 45) x 100% = 18.2 % W/W *(include this term in your answer)*

Units of Concentration

Example 1:

What is the percent by volume concentration of a solution in which 75.0 ml of ethanol is diluted to a volume of 250.0 ml?

$$\% \text{ ethanol} = \frac{75.0 \text{ ml}}{250.0 \text{ ml}} \times 100$$

$$= 30.0\%$$

Units of Concentration

Example 2:

What volume of acetic acid is present in a bottle containing 350.0 ml of a solution which measures 5.00% concentration?

$$\frac{V_{ethanol}}{350.0 \text{ ml}} = 0.05$$

x = 17.5 ml

Units of Concentration

Example 3:

Find the percent by mass in which 41.0 g of NaCl is dissolved in 331 grams of water.

$$\frac{41\text{ g}}{372\text{ g}} \times 100 = 11.0\%$$

Extremely Low Concentrations

- In some instances molar concentration is not an effective measuring tool, such as when concentrations are very low. Example: Environmental pollutants.

- In these situations, scientists use:
  - ppm: parts per million, $1:10^6$ (drop in a full bathtub)
  - ppb: parts per billion, $1:10^9$ (drop in a large swimming pool)
  - ppt: parts per trillion $1:10^{12}$ (drop in 1000 swimming pools)

- These are a special case of %W/W concentration.

Extremely small concentrations

- Statistics are most often described in this manner; for example mortality rates are described in deaths per 100,000 people. (Canada's mortality rate is 7.74; USA's is 8.38).

Picture it!

Measure ment	Equivalent Ratio
1 ppm	• 1 drop of water in a full bathtub • 1 second in 11.5 days • 1 minute in 2 years
1 ppb	• 1 drop of water in a full swimming pool • 1 square of toilet paper in a line stretching from New York to London, England • 1 second in 32 years • 1 pinch of salt in 10160.5kg of potato chips

Calculating small concentrations

- In doing quantitative calculations, as we must first ensure that the **units** of the solute and solvent are the same before working out the stoichiometry.
-
- **Example 1.**
- In Ontario, drinking water should have less than 10 ppb of lead in it. What is the maximum amount of lead allowed in a 500 mL bottle of water?
-
- **1mL of water = 1g of water (using density)**
- Therefore 10ppb = 10g in 1,000,000,000g of water

- Set up equivalent ratios:

$$\frac{10g}{1 \times 10^{9}g} = \frac{x}{500g}$$

$$x = 5 \times 10^{-6}g = 5\mu g$$

Know your prefixes. See the table on the next page.

The World of Metric Prefixes We Sort Of Care About [(1)]

Now with mostly useless footnotes!

Prefix	Abbreviation	What they mean (human terms)	What they mean (numerically)	What they mean (scientific notation)	An example
mega	M	million	1,000,000	10^6	There are a million bytes i
kilo	k	thousand	1,000	10^3	There are a thousand me
deci	d	tenth	0.1	10^{-1}	There is a tenth of a liter i
centi	c	hundredth	0.01	10^{-2}	There is a hundredth of a
milli	m	thousandth	0.001	10^{-3}	There is a thousandth of a
micro	μ [(2)]	millionth	0.000001	10^{-6}	There is a millionth of a m
nano	n	billionth	0.000000001	10^{-9}	There is a billionth of a m

Example 2

- A 30 g sample of herbal medicine was recently banned because it was found to contain 0.0705 mg of strychnine (a powerful naturally occurring poison). What was this toxic level in ppm?
-
- 1 ppm = 1 g in 1,000,000g sample = 1×10^6 g
- 0.0705mg = 7.05×10^{-5}g NOTE: MUST know your metric conversions

$$\frac{7.05 \times 10^{-5} g}{30 g} = \frac{x}{1 \times 10^{6} g}$$

$$x = 2.35 g$$

- Therefore, there is 2.35g of toxin per million grams of medicine, or 2.35ppm

END HERE

Making Solutions

- You just calculated the molar mass of sodium chloride to be 58.44 g/mol.
- To determine how to make a stock solution of sodium chloride, use the formula:

$$g = M \times L \times \text{molar mass}$$

Making Solutions

- How many grams of NaCl would you need to prepare 200.0 mL of a 5 M solution?

$$g = M \times L \times \text{molar mass}$$

$$g = (5\text{mol/L})\ (0.2\text{L})\ (58.44\text{g/mol})$$

$$g = 58.44\ g$$